Appendices to "Explaining Feature Contributions in Multimodal Instagram Engagement Prediction"

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A Experimental Details

Preprocessing. Posts are restricted to 2017–2018 observations with English captions, at least one valid image, and a valid metadata–image mapping. Captions are cleaned by removing hyperlinks, email addresses, and user mentions, while hashtags are reduced to their textual component. For TF–IDF, stopwords are also removed and the vocabulary is fitted on training data only. Images are converted to RGB and resized to 224×224. For posts with multiple images, image-level representations are mean-pooled into one post-level vector. Categorical metadata variables are encoded as numerical indicators and numerical variables are standardized. All data-dependent steps are fitted on training data and applied unchanged to validation and test data.

Split and balancing protocol. A chronological split avoids mixing past and future posts: training covers January 2017–October 2018, validation November 2018, and test December 2018. Validation is used for model selection and hyperparameter tuning, while test data are held out for final reporting. Category balancing is applied only to training. The smallest creator category, *pet*, contains 544 influencers and defines the reference size. For each other category, 544 influencers are selected through a reproducible pseudo-random procedure based on sorted username hashes, retaining all their posts. Validation and test remain unchanged, preserving their natural distributions.

Table 1: Metadata feature groups used in the experiments.

Group	**Variables**
Temporal	`year`, `month`, `dow`, `hour_utc`
Caption structure	`has_caption`, `caption_len_char`, `n_hashtags`, `n_mentions`, `n_urls`, `n_emojis`
Image geometry	`width`, `height`, `aspect_ratio`, `area`, `orientation`
Creator category	`category`
Account-level structural	`followees`, `posts`
Excluded target-related	`like_count`, `comment_count`, `followers`, `engagement_rate`

Table 2: Classifier-level macro-F1 by modality and representation; bold marks row maxima.

Family	Configuration	Linear SVM	NB	RF	XGBoost
Traditional	Text, binary	**0.61**	0.54	0.54	0.55
Traditional	Text, 3-class	**0.43**	0.36	0.29	0.35
Traditional	Text, 5-class	**0.29**	0.22	0.17	0.22
Traditional	Image, binary	0.55	0.49	0.55	**0.57**
Traditional	Image, 3-class	0.38	0.30	0.38	**0.39**
Traditional	Image, 5-class	0.23	0.17	0.23	**0.25**
Traditional	Multimodal, binary	0.68	0.58	0.67	**0.69**
Traditional	Multimodal, 3-class	0.47	0.40	0.49	**0.51**
Traditional	Multimodal, 5-class	0.23	0.23	0.32	**0.33**
SBERT + EfficientNet	Text, binary	0.57	0.55	0.57	**0.58**
SBERT + EfficientNet	Text, 3-class	0.38	0.37	0.39	**0.40**
SBERT + EfficientNet	Text, 5-class	0.24	0.22	0.23	**0.25**
SBERT + EfficientNet	Image, binary	**0.59**	0.57	0.58	**0.59**
SBERT + EfficientNet	Image, 3-class	**0.41**	0.37	**0.41**	**0.41**
SBERT + EfficientNet	Image, 5-class	0.25	0.21	0.25	**0.26**
SBERT + EfficientNet	Multimodal, binary	**0.60**	0.58	0.59	**0.60**
SBERT + EfficientNet	Multimodal, 3-class	0.41	0.38	0.41	**0.42**
SBERT + EfficientNet	Multimodal, 5-class	0.25	0.21	0.23	**0.26**
CLIP	Text, binary	**0.58**	0.55	0.57	**0.58**
CLIP	Text, 3-class	**0.41**	0.37	0.38	**0.41**
CLIP	Text, 5-class	**0.25**	0.21	0.22	**0.25**
CLIP	Image, binary	**0.62**	0.60	0.61	**0.62**
CLIP	Image, 3-class	0.42	0.40	0.43	**0.44**
CLIP	Image, 5-class	0.16	0.08	**0.19**	0.17
CLIP	Multimodal, binary	0.68	0.63	0.67	**0.69**
CLIP	Multimodal, 3-class	0.45	0.44	0.50	**0.52**
CLIP	Multimodal, 5-class	0.30	0.26	0.32	**0.34**

B Feature Details

Metadata features. Table 1 reports the metadata feature groups used in the experiments. The predictive feature set includes temporal, caption-structure, image-geometry, creator-category, and account-level structural variables. Target-related variables are excluded to avoid direct leakage from the engagement-rate definition, i.e., `like_count`, `comment_count`, `followers`, and `engagement_rate`.

Text and image representations. The traditional family represents captions with TF–IDF unigram vectors, using `min_df`= 20, `max_df`= 0.7, a maximum of 50,000 features, and category-specific top-word indicators. Its image view is represented by 149 handcrafted descriptors, including colorfulness, Hue Saturation Lightness statistics and histograms, grayscale histograms, entropy, Laplacian variance, edge density, Local Binary Patterns, and Gray-Level Co-Occurrence Matrix features. The semantic family uses 384-dimensional SBERT caption embeddings and 1280-dimensional EfficientNet-B0 image embeddings. The vision–language family uses CLIP text and image embeddings, both with 512 dimensions.

C Model Details

The experiments compare four classifier families: Linear SVM as a linear margin-based classifier, Gaussian Naive Bayes as a probabilistic baseline, Random Forest as a bagging-based ensemble, and XGBoost as a gradient-boosted tree model. Only the hyperparameters required for model selection are tuned, keeping the validation protocol computationally bounded and consistent across representation families. Table 2 reports classifier-level test macro-F1 for the configurations used in Table 3 of the main paper. Its multimodal rows support the comparison across representation families, while its text-only, image-only, and multimodal rows together support the within-family comparison across modality settings.